Component	Description/function	Significance
	using radio or satellite communication.	mission data for enhanced situational awareness.
Power source (battery or fuel tank)	Provided electrical or fuel energy required for propulsion, avionics, and onboard systems.	Determined overall flight endurance, payload capacity, and mission duration.
Payload module	Carried modular equipment such as cameras, infrared sensors, or scientific instruments for mission-specific applications.	Enabled flexible, multifunctional use in surveillance, reconnaissance, mapping, and environmental research.
Ground control station (GCS)	Operator interface that displayed real-time flight data, mission parameters, and video feeds.	Allowed mission planning, monitoring, route adjustments, and overall UAV coordination.

3.3.3 System integration and functional operation

In operation, the ground control station (GCS) transmits control signals to the UAV via radio or satellite link, commanding throttle, altitude, and navigation. The flight controller unit (FCU) interprets these inputs and adjusts the propulsion and control surfaces based on feedback from the IMU and GPS systems. Data from onboard cameras and sensors is transmitted back to the GCS for real-time analysis. This closed-loop system enabled semi-autonomous mission execution with minimal manual intervention, greatly improving operational safety and efficiency.

The integration of electronic control systems, navigation aids, and modular payloads marked a major technological leap over earlier UAV generations. These systems not only improved flight performance but also